

Statistical Physics PHYS512

HW #1

Fall 2026

1. Explain the ideal gas system and the absolute temperature defined in thermodynamics. Find the answer in your study with reference books.
2. Calculate the heat capacity of an ideal gas at a constant pressure $C_p = \frac{dQ}{dT}|_p$. [Hint: In this case, the gas can expand against the pressure p when absorbing the heat, which means $dQ = dE - dW$ according to the 1st law of thermodynamics. Thus, $C_p(T) = C_v(T) + \frac{d(-W)}{dT}|_p$.]
3. Explain the zeroth law of thermodynamics.
4. A differential of the form

$$df = P(x, y)dx + Q(x, y)dy$$

is exact differential if

$$df = \frac{\partial f}{\partial x}dx + \frac{\partial f}{\partial y}dy$$

and thus $\int df$ is path-independent. Find the relation for P and Q for making df is an exact differential.

[Note: If you are not familiar with exact differential, you solve this problem after your self-study.]